

gdTAI V4.2 NK and T lineage cluster review

This read-only review uses the completed development-only scVI representation. UMAP coordinates and marker plots use the deterministic 250,000-cell diagnostic sample; cluster cell and anchor counts use all 4,023,462 cells. No locked cohort was included.

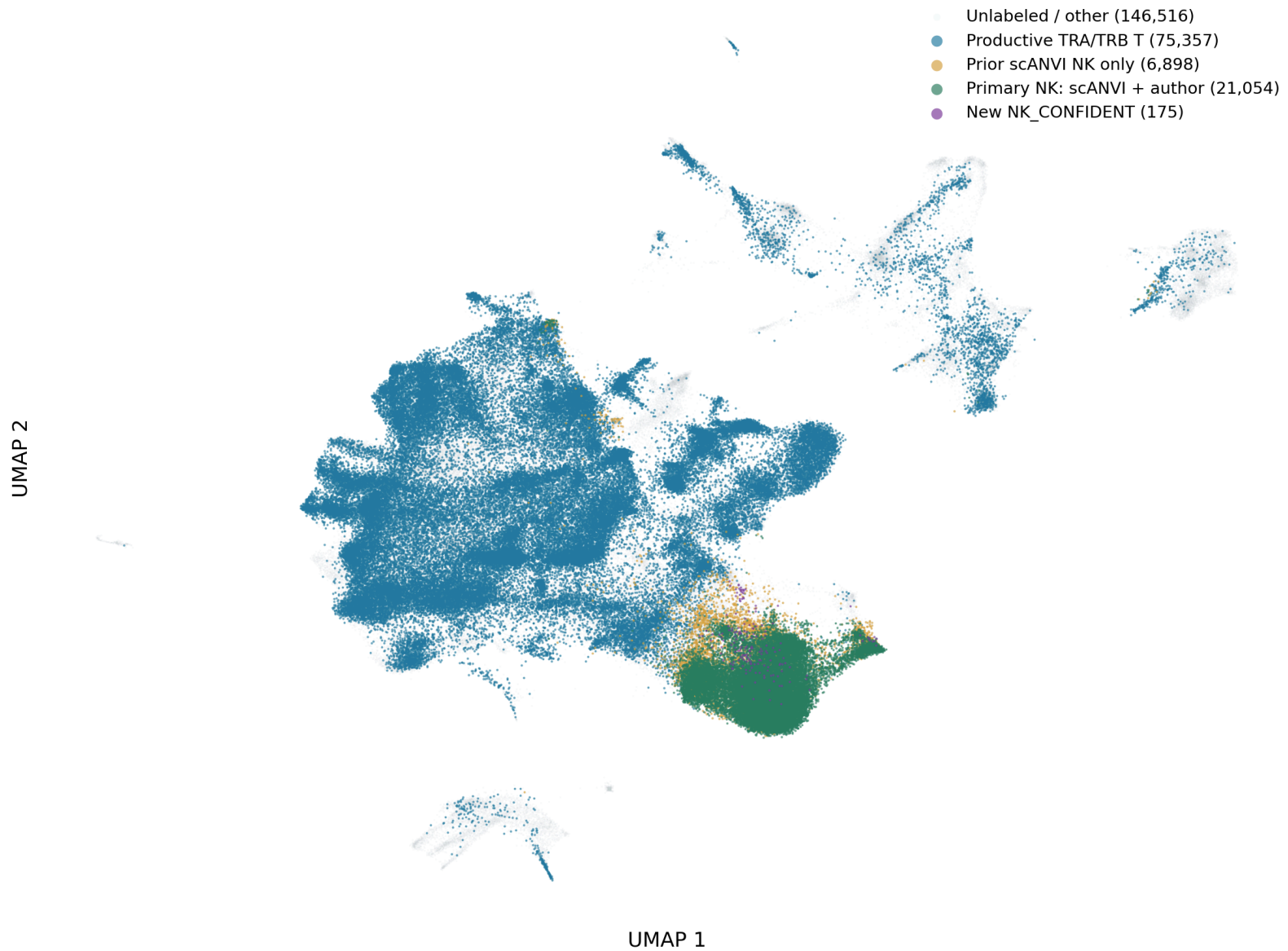
Result: PASS_VISUAL_REVIEW_EXTENDED. The feature plots, recoverable annotation evidence, independent Leiden clusters, and exact raw CD3/TCR detection counts are ready for biological review before changing the NK label definition.

24 unsupervised clusters	21,054 primary NK anchors	967,149 restored productive-T anchors	469 current NK_CONFIDENT cells
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1. Annotation evidence on the scVI UMAP

The missing historical integrated H5AD prevents recovery of its complete full-cell scANVI annotation column. The plot therefore shows only traceable evidence: prior scANVI-NK cells retained in the frozen V4 manifest, dual scANVI-plus-author NK anchors, restored productive-TRA/TRB T anchors, and the 469 current two-evidence NK candidates.

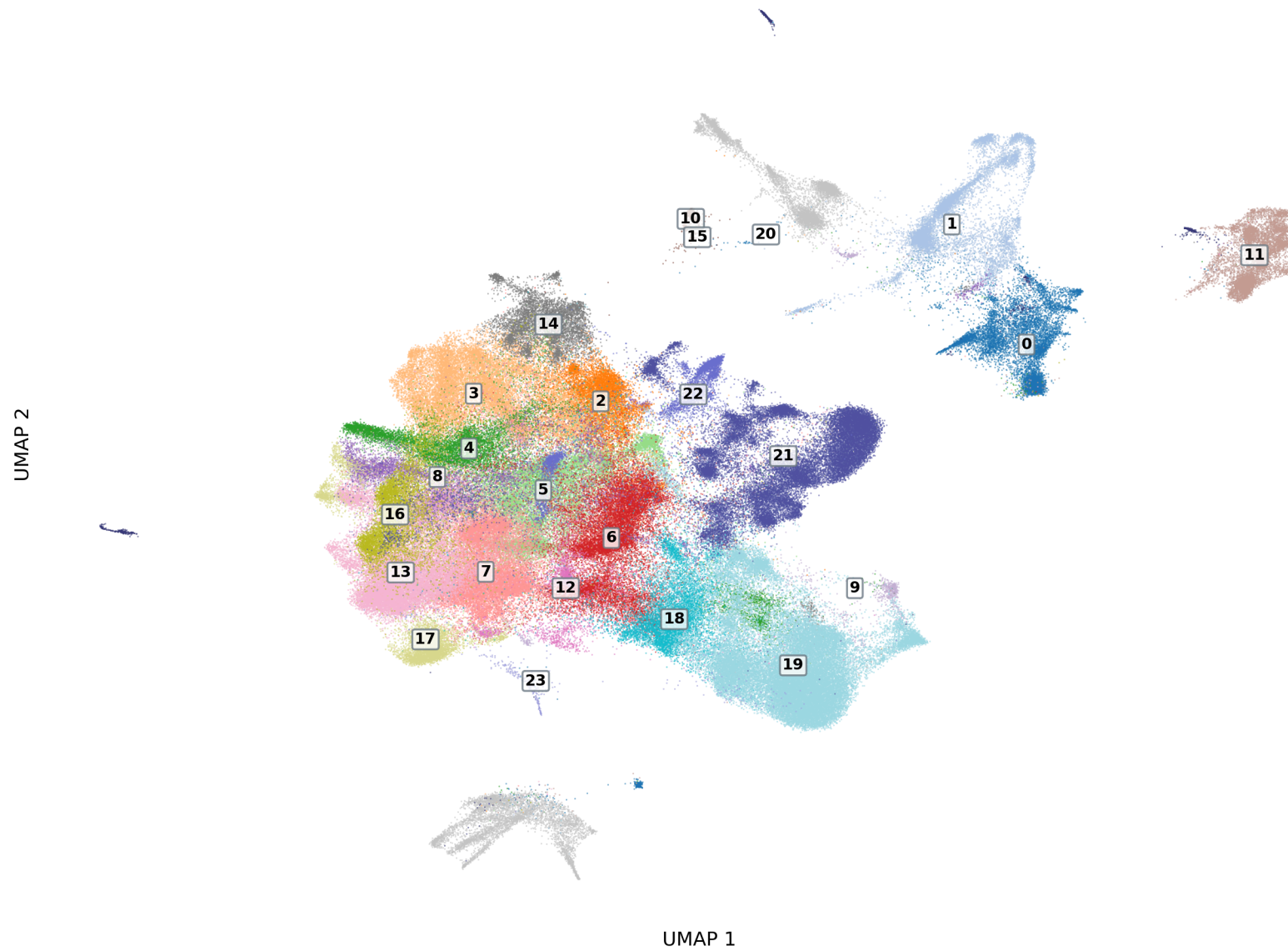
Recoverable prior annotation evidence on the scVI UMAP



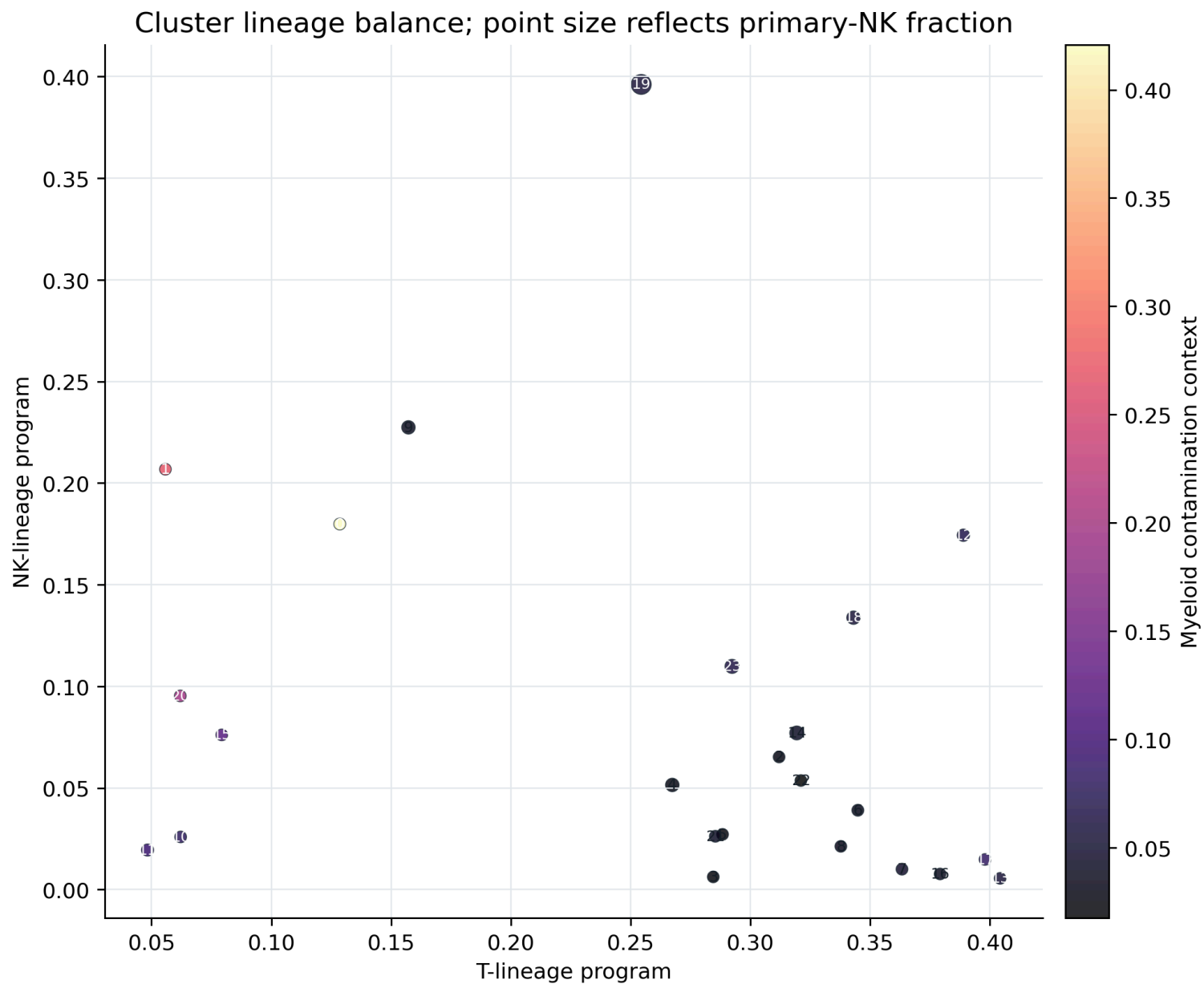
These are provenance classes, not a new annotation model.

2. Unsupervised structure

Unsupervised scVI Leiden clusters: global_r0.8_s29



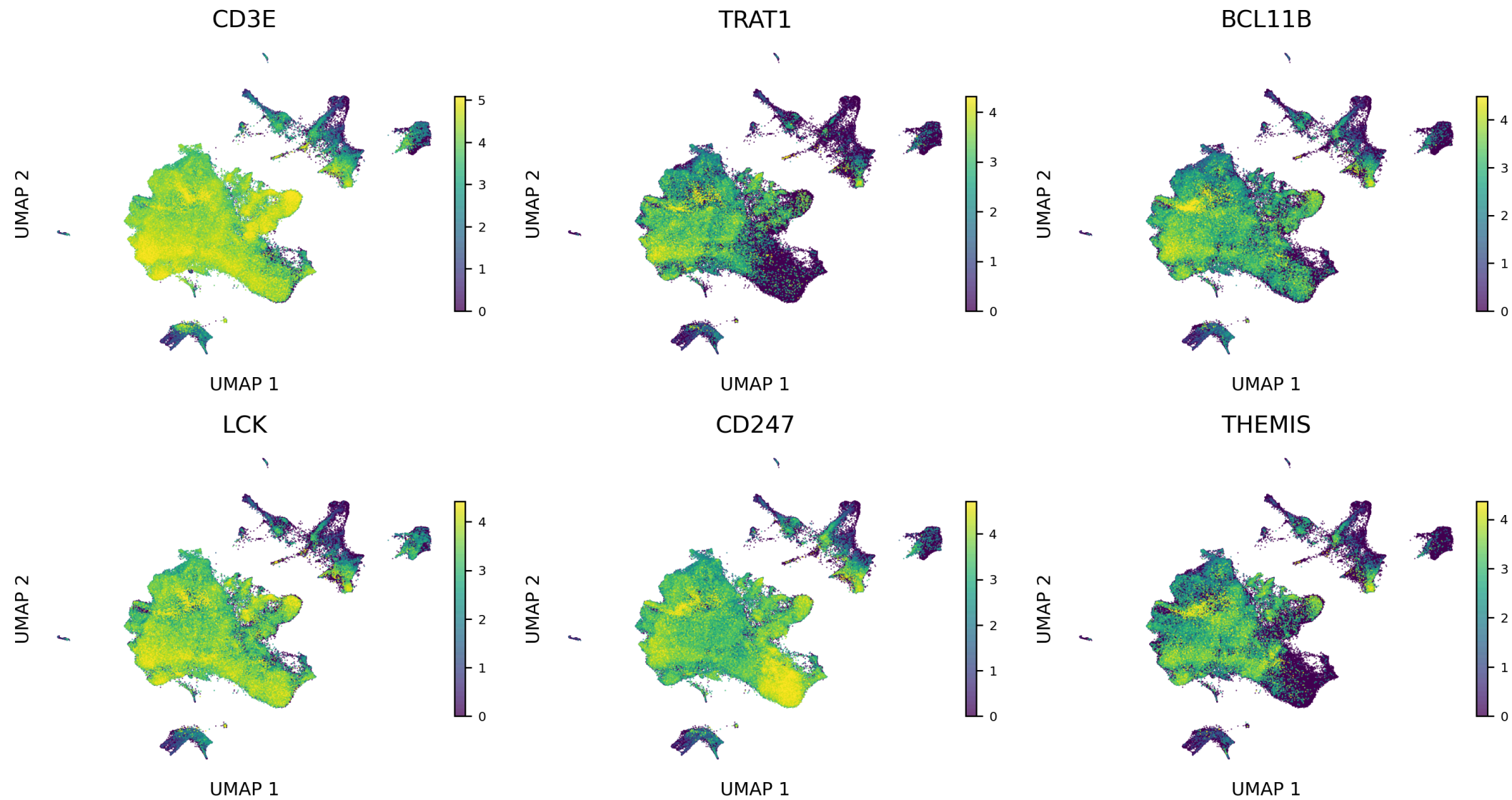
Frozen global Leiden partition `global_r0.8_s29`.



Scores combine multiple genes; color shows myeloid contamination context and point size shows primary-NK fraction.

3. T-cell lineage feature plots

T-cell lineage markers on the scVI UMAP

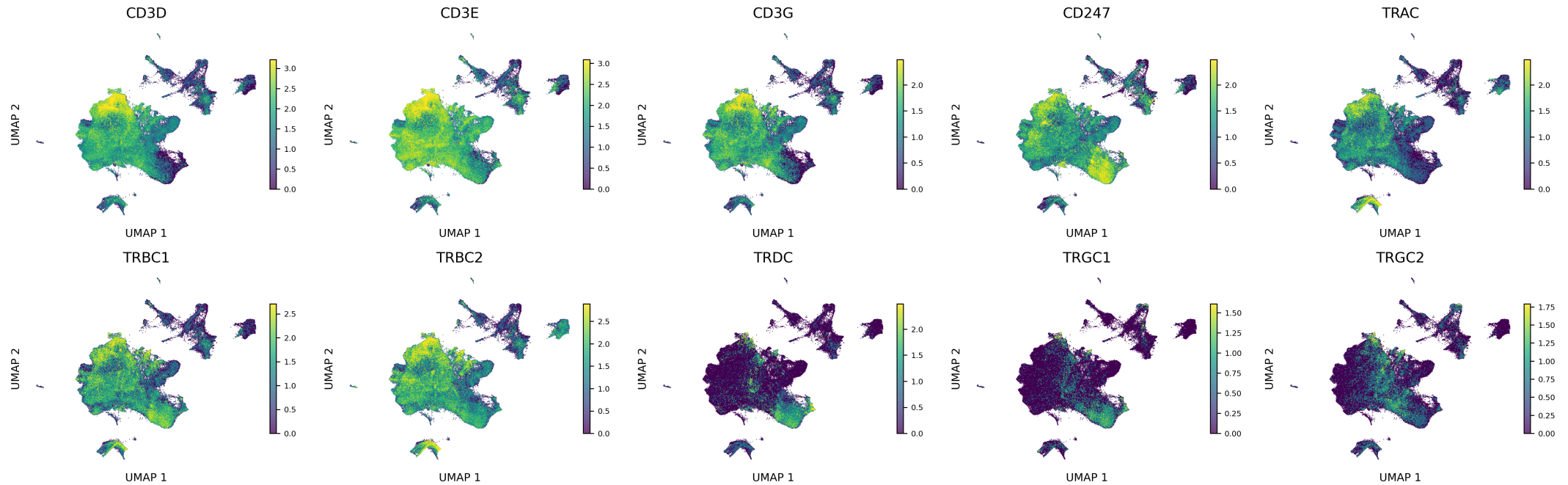


Expression is log1p CP10K within the fixed 4,000 scVI integration features.

4. Additional CD3 and TCR genes

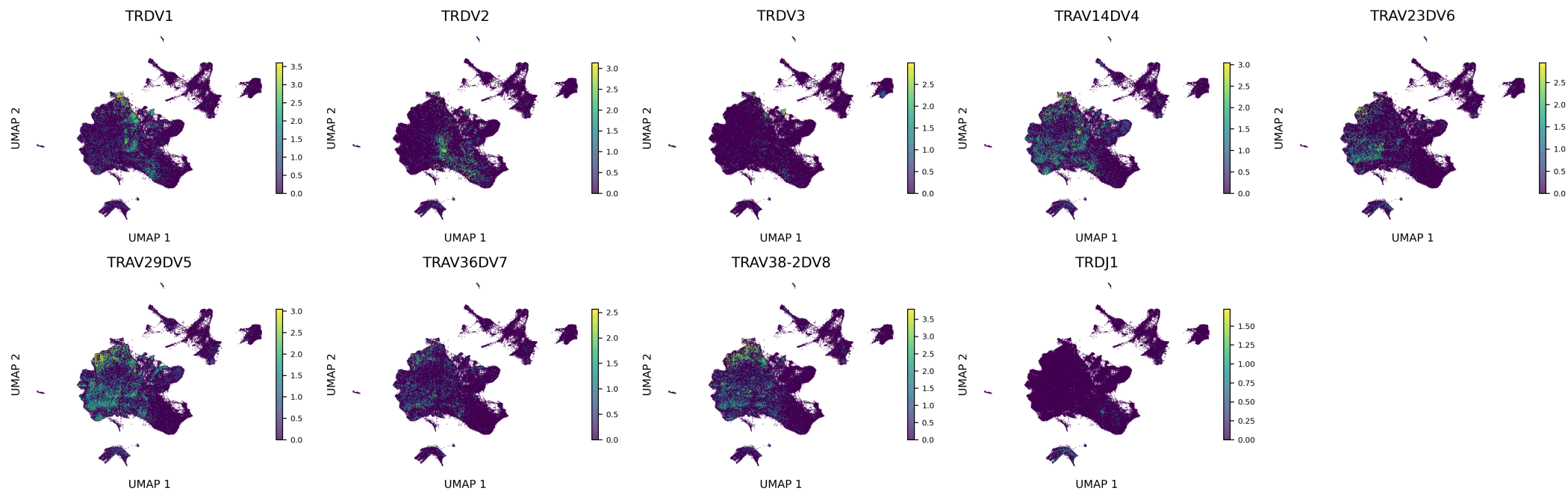
Expression source matters. The frozen 4,000-gene scVI matrix contains only CD3E and CD247 from this panel because TCR genes were deliberately excluded from HVG selection. The plots below therefore overlay $\log_{1p}(\text{raw UMI})$ from the six registered source H5ADs on the unchanged scVI UMAP. Exact counts use all 4,023,462 cells, not only the visualization sample, and all 6 source H5ADs contain all 29 displayed genes.

CD3 complex and TCR constant-chain genes on the scVI UMAP



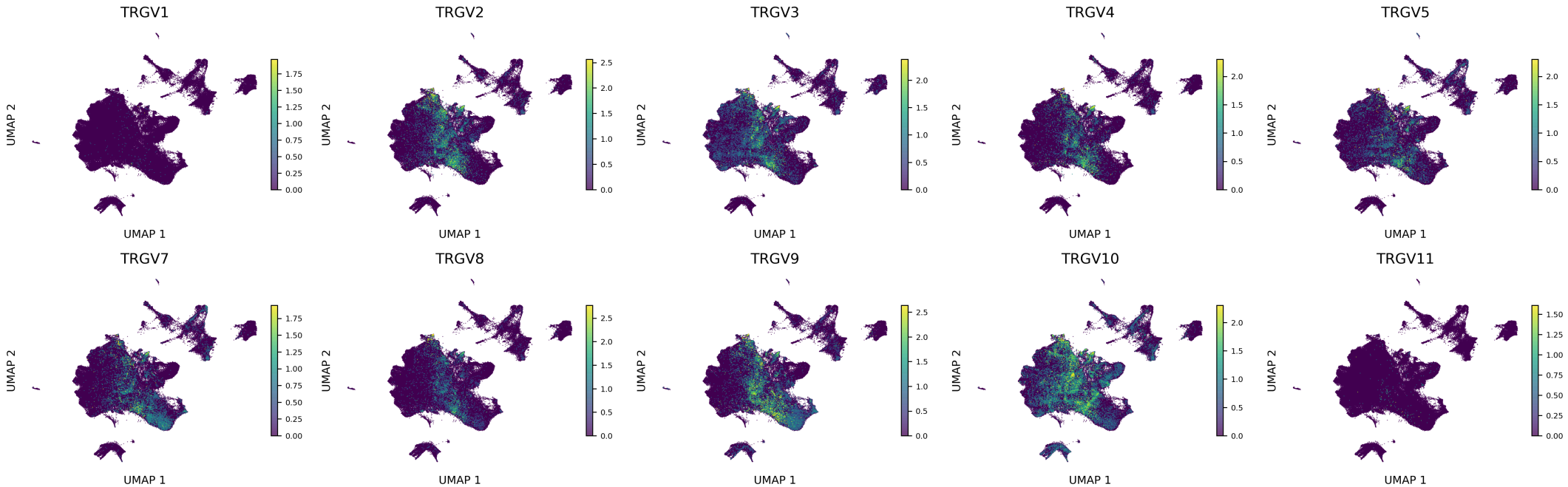
CD3 complex and TCR constant-chain raw expression on the scVI UMAP.

TCR delta V/J genes on the scVI UMAP



Delta V/J genes, including the delta-associated V genes embedded in the TRA locus.

TCR gamma V genes on the scVI UMAP



Gamma V expression is shown for context and is not treated as uniquely gamma-delta specific.

Across the full sidecar, **3,435,095** cells express at least one of CD3D/E/G, **290,288** express TRDC, and **652,723** express at least one selected delta-V gene. There are **222,616** TRDC-positive cells without detected delta-V expression, consistent with either dropout or non-specific/ambient constant-chain signal depending on the surrounding lineage evidence.

Population	Cells	CD3D/E/G+ n	CD3D/E/G+ %	TRA/B constant+ n	TRA/B constant+ %	TRDC+ n	TRDC+ %	Delta-V+ n	Delta-V+ %	TRDC+ Delta-V- n	TRDC+ Delta-V- %	Gamma-V+ n	Gamma-V+ %
All development sidecar	4,023,462	3,435,095	85.38%	3,058,977	76.03%	290,288	7.21%	652,723	16.22%	222,616	5.53%	1,086,515	27.00%
Primary NK anchors	21,054	8,242	39.15%	17,105	81.24%	8,487	40.31%	396	1.88%	8,277	39.31%	4,642	22.05%
Productive TRA/TRB T anchors	967,149	957,169	98.97%	819,957	84.78%	16,946	1.75%	186,121	19.24%	11,093	1.15%	244,243	25.25%
All eligible candidates	113,287	78,528	69.32%	58,112	51.30%	6,161	5.44%	11,668	10.30%	3,789	3.34%	23,040	20.34%
Cluster 19 all cells	490,300	295,409	60.25%	418,329	85.32%	181,765	37.07%	51,030	10.41%	161,663	32.97%	197,372	40.26%
Cluster 19 eligible candidates	9,162	4,353	47.51%	6,678	72.89%	1,952	21.31%	451	4.92%	1,784	19.47%	3,473	37.91%
Strict NK core (cluster 19 both evidence)	461	0	0.00%	350	75.92%	152	32.97%	2	0.43%	152	32.97%	195	42.30%
Current confident outside cluster 19	8	0	0.00%	3	37.50%	0	0.00%	0	0.00%	0	0.00%	3	37.50%
Cluster 9 all cells	41,060	16,894	41.14%	29,634	72.17%	17,834	43.43%	1,312	3.20%	17,200	41.89%	12,532	30.52%
Cluster 9 eligible candidates	1,139	485	42.58%	630	55.31%	303	26.60%	29	2.55%	292	25.64%	306	26.87%

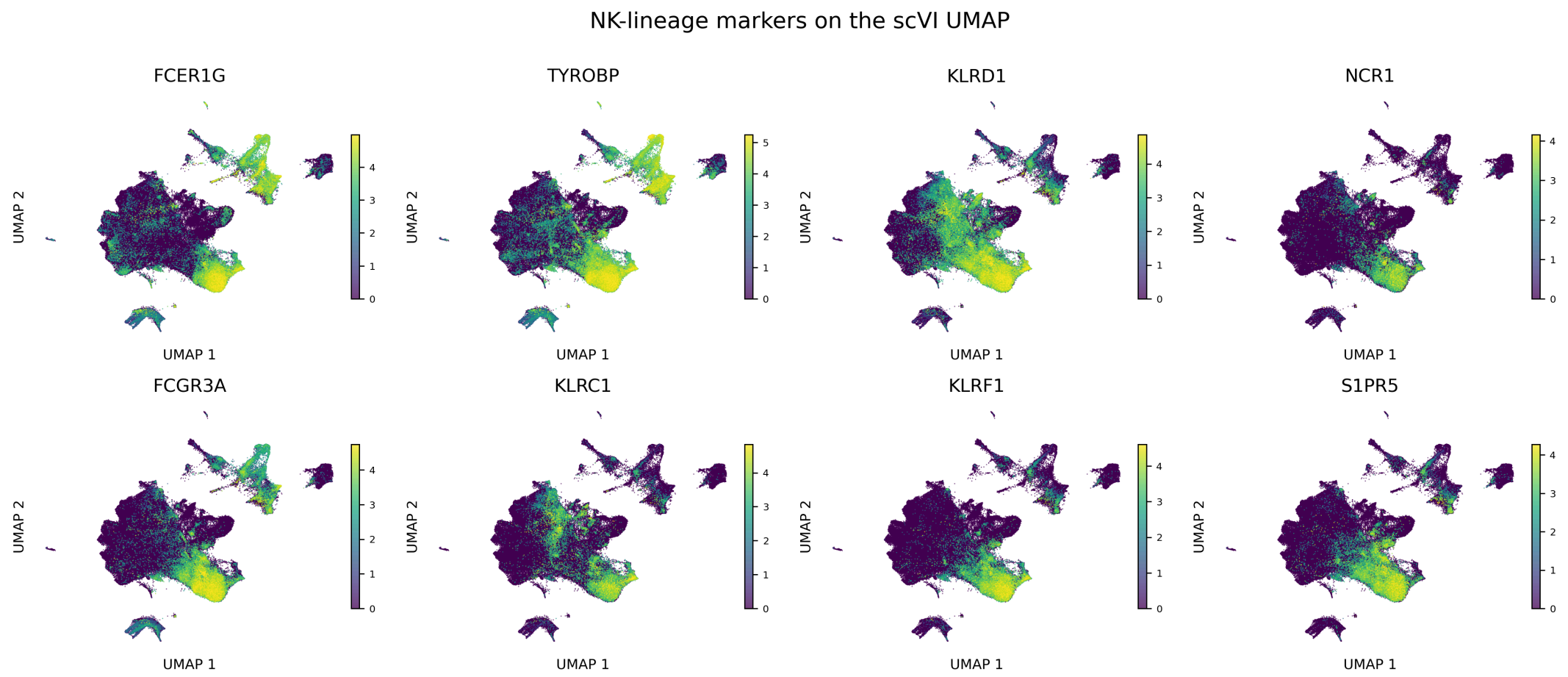
Gene	Panel	Detected cells	Detected %
CD3D	CD3_complex	2,662,995	66.19%
CD3E	CD3_complex	3,170,723	78.81%
CD3G	CD3_complex	2,112,641	52.51%
CD247	CD3_complex	2,271,048	56.45%
TRAC	TCR_constant	1,193,424	29.66%
TRBC1	TCR_constant	1,524,836	37.90%
TRBC2	TCR_constant	2,320,383	57.67%
TRDC	TCR_constant	290,288	7.21%
TRGC1	TCR_constant	187,532	4.66%
TRGC2	TCR_constant	271,270	6.74%
TRDV1	TCR_delta_VJ	90,907	2.26%
TRDV2	TCR_delta_VJ	59,417	1.48%
TRDV3	TCR_delta_VJ	24,261	0.60%
TRAV14DV4	TCR_delta_VJ	130,749	3.25%
TRAV23DV6	TCR_delta_VJ	73,328	1.82%
TRAV29DV5	TCR_delta_VJ	163,183	4.06%
TRAV36DV7	TCR_delta_VJ	56,332	1.40%
TRAV38-2DV8	TCR_delta_VJ	101,094	2.51%
TRDJ1	TCR_delta_VJ	5,234	0.13%
TRGV1	TCR_gamma_V	2,862	0.07%
TRGV2	TCR_gamma_V	194,343	4.83%
TRGV3	TCR_gamma_V	205,955	5.12%
TRGV4	TCR_gamma_V	127,277	3.16%
TRGV5	TCR_gamma_V	130,784	3.25%
TRGV7	TCR_gamma_V	137,135	3.41%
TRGV8	TCR_gamma_V	102,396	2.54%
TRGV9	TCR_gamma_V	227,775	5.66%
TRGV10	TCR_gamma_V	292,149	7.26%
TRGV11	TCR_gamma_V	4,074	0.10%

The 461-cell core is not yet clean enough to freeze as NK truth. Zero CD3D/E/G expression is circular because absence of those genes was part of the strict expression rule. In contrast, 350 cells (75.9%) have at least one TRA/TRB constant-chain UMI, 250 (54.2%) have at least two,

and 178 (38.6%) have at least three. This is too frequent to dismiss as universal single-UMI ambient signal without a source- and library-resolved ambient/doublet audit.

Gene group	Minimum total UMI	Cells	Fraction of 461	Median UMI among positive	P90 UMI among positive
CD3D_E_G	1	0	0.00%	0.0	0.0
CD3D_E_G	2	0	0.00%	0.0	0.0
CD3D_E_G	3	0	0.00%	0.0	0.0
CD3D_E_G	5	0	0.00%	0.0	0.0
alpha_beta_constant	1	350	75.92%	3.0	7.0
alpha_beta_constant	2	250	54.23%	3.0	7.0
alpha_beta_constant	3	178	38.61%	3.0	7.0
alpha_beta_constant	5	80	17.35%	3.0	7.0
gamma_delta_constant	1	186	40.35%	2.0	4.0
gamma_delta_constant	2	95	20.61%	2.0	4.0
gamma_delta_constant	3	43	9.33%	2.0	4.0
gamma_delta_constant	5	15	3.25%	2.0	4.0
delta_V	1	2	0.43%	1.0	1.0
delta_V	2	0	0.00%	1.0	1.0
delta_V	3	0	0.00%	1.0	1.0
delta_V	5	0	0.00%	1.0	1.0
gamma_V	1	195	42.30%	1.0	2.0
gamma_V	2	56	12.15%	1.0	2.0
gamma_V	3	14	3.04%	1.0	2.0
gamma_V	5	1	0.22%	1.0	2.0

5. NK lineage feature plots

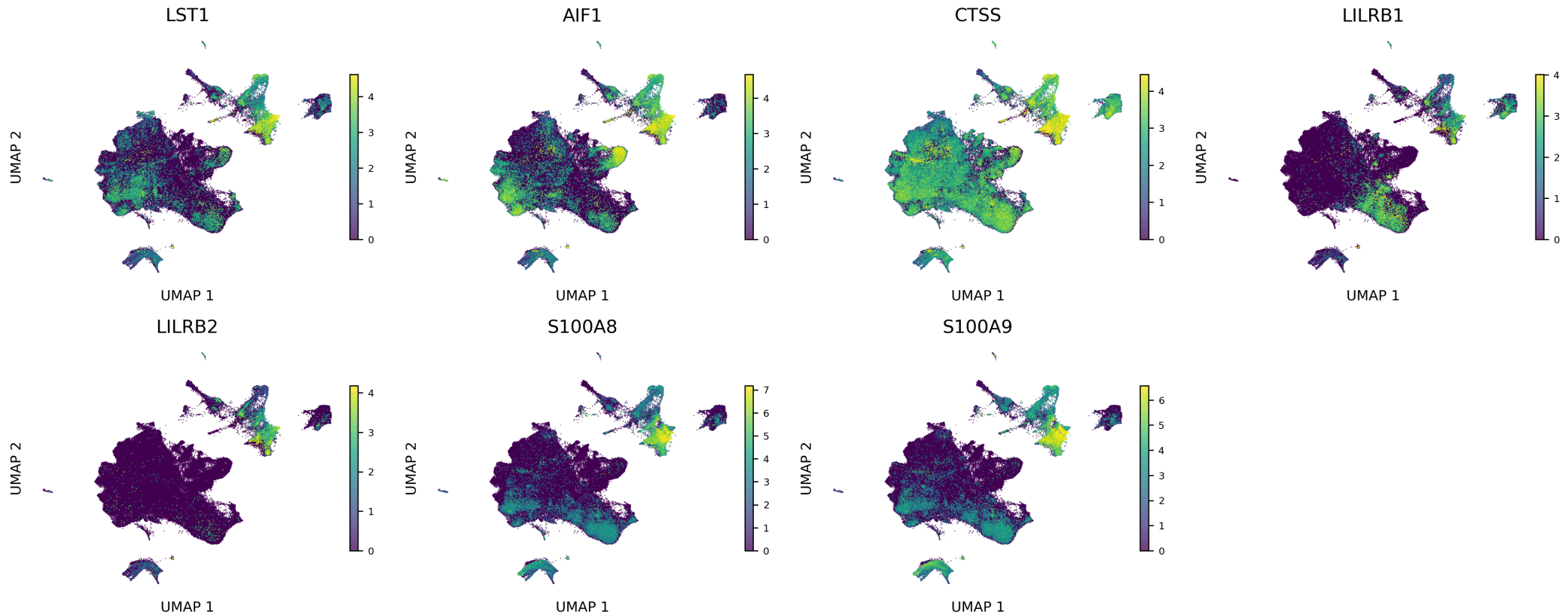


NK evidence requires a multigene program; no single receptor is treated as definitive.

6. Non-T/NK contamination context

FCER1G and TYROBP are not sufficient for NK identity because myeloid cells also express them. The following panel is used only to recognize obvious non-T/NK clusters; these genes are not proposed as gdTAI classifier features.

Myeloid contamination context on the scVI UMAP

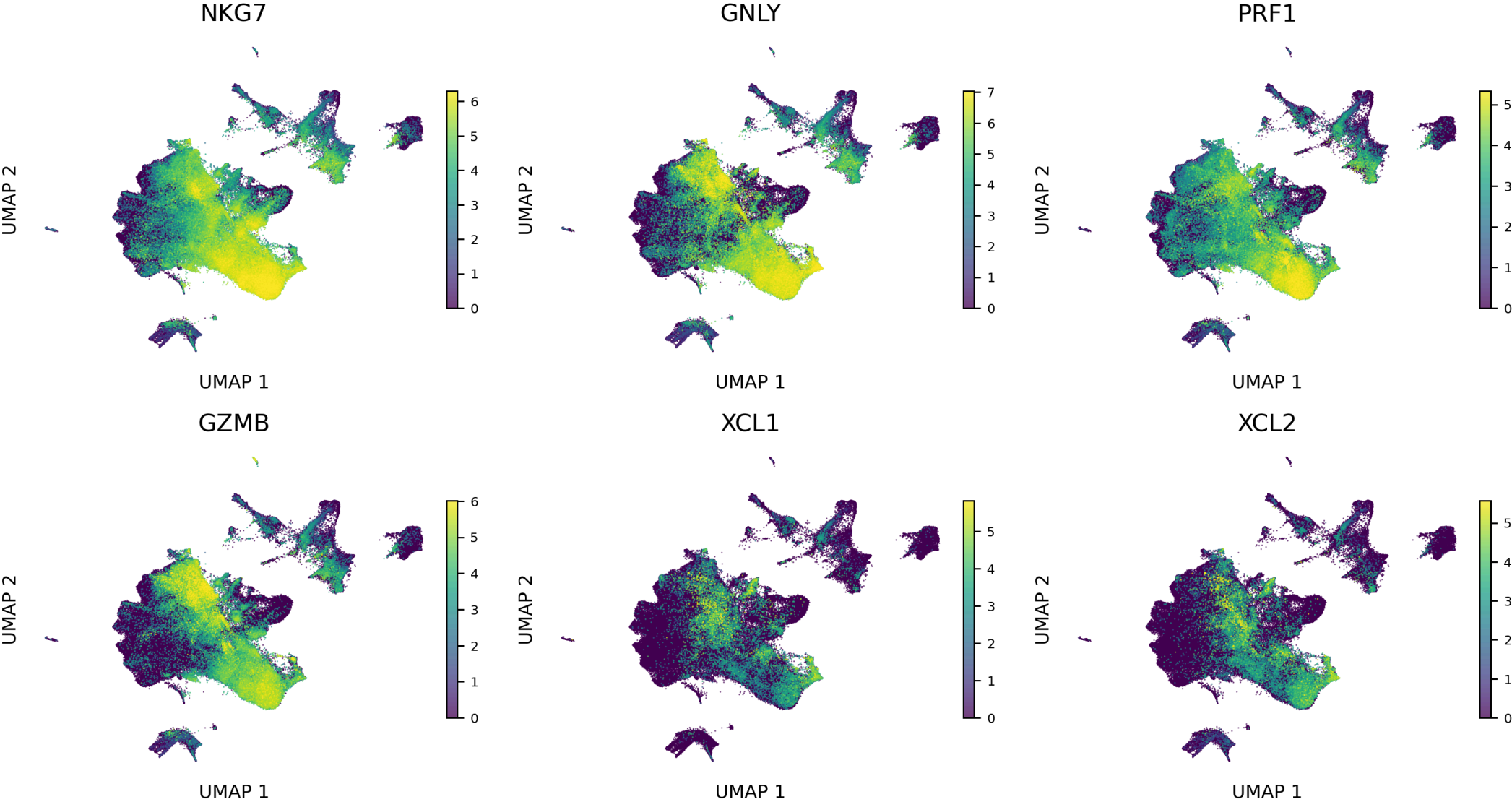


Clusters with broad LST1/AIF1/CTSS/LILRB/S100 support should not be called NK solely from adaptor expression.

7. Shared cytotoxic program

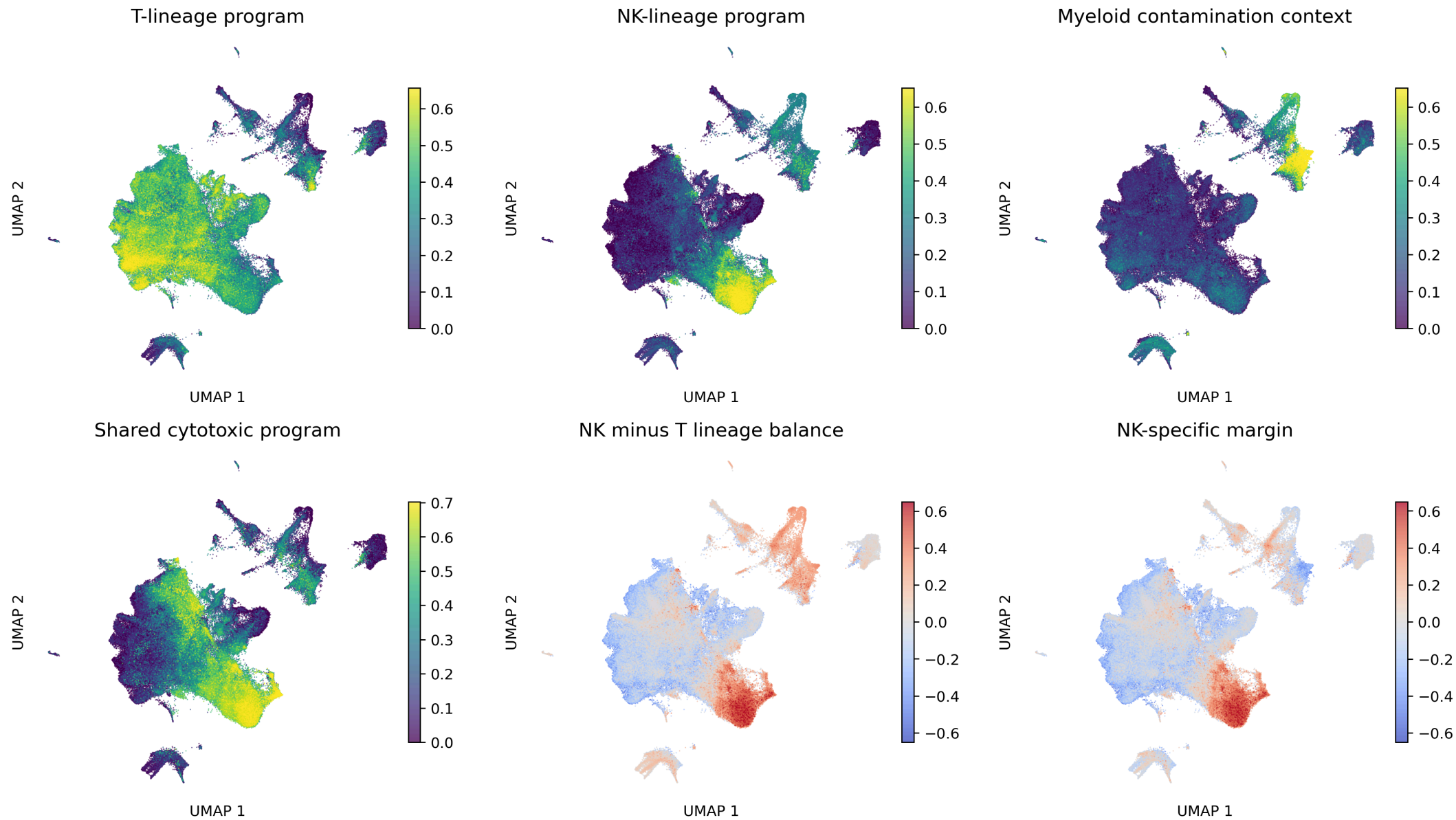
Do not use cytotoxicity as NK truth. NKG7, GNLY, PRF1, GZMB, XCL1, and XCL2 can also be high in CD8 and gamma-delta T cells.

Shared cytotoxic markers on the scVI UMAP



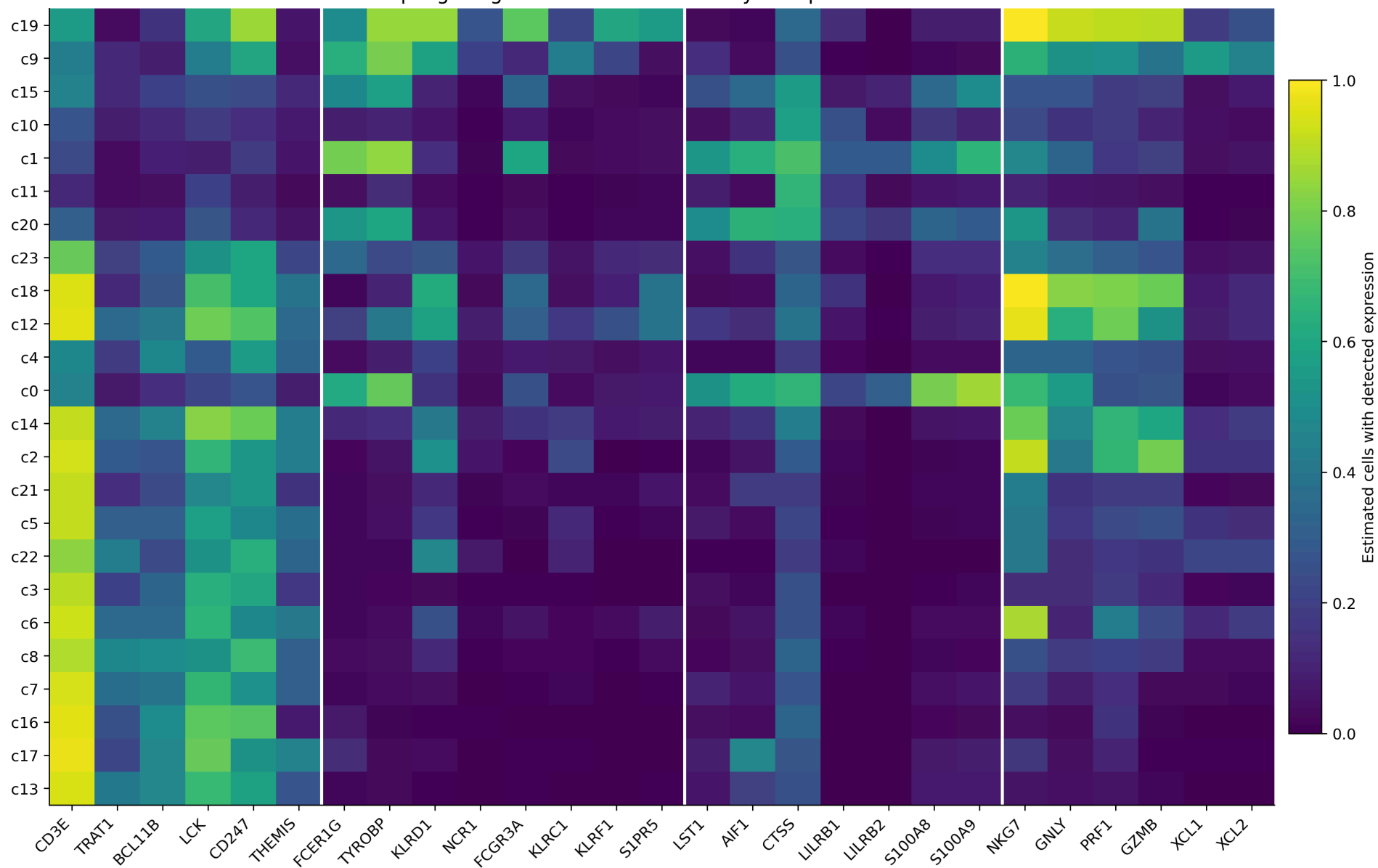
These markers are shown as phenotype context only.

8. Multigene programs and cluster heatmap



Each program is the mean of gene-wise robust-scaled expression. NK-specific margin subtracts the stronger of T-lineage or myeloid-context evidence.

Sampling-weighted marker detection by unsupervised cluster



Sampling-weighted marker detection estimates compensate for the diagnostic sample's source and evidence-role oversampling.

9. Clusters ranked for review

The table ranks clusters by the continuous NK-specific margin: NK lineage minus the stronger of T-lineage or myeloid-context evidence. It does not impose a new hard NK cutoff. Exact anchor, candidate, and source-composition counts are included so expression, annotation provenance, and source bias can be considered together.

cluster	nk_likeness_rank	n_cells	n_primary_nk	n_productive_t	n_eligible_candidates	n_confident_nk	weighted_t_program	weighted_nk_program	weighted_myeloid_program	weighted_cytotoxic_program	nk_minus_t_program	nk_specific_margin	dominant_candidate_source	dominant_candidate_source_fraction
19	1	490300	20237	18316	9162	461	0.254	0.396	0.054	0.522	0.142	0.142	GSE292700	86.7%
9	2	41060	48	686	1139	1	0.157	0.227	0.037	0.309	0.070	0.070	GSE294273	49.3%
15	3	39722	0	2829	14043	0	0.079	0.076	0.119	0.050	-0.003	-0.043	GSE292700	80.2%
10	4	9276	0	907	181	0	0.062	0.026	0.079	0.054	-0.036	-0.053	GSE292700	66.9%
1	5	42063	0	1554	16940	0	0.056	0.207	0.269	0.086	0.151	-0.062	GSE292700	96.9%
11	6	112032	8	5570	5772	0	0.048	0.019	0.093	0.025	-0.029	-0.073	GSE294273	85.8%
20	7	7922	0	515	488	0	0.062	0.095	0.196	0.091	0.033	-0.100	GSE294273	88.3%
23	8	11226	26	4894	39	0	0.292	0.110	0.062	0.168	-0.183	-0.183	GSE292700	97.4%
18	9	236832	262	80173	668	0	0.343	0.134	0.056	0.425	-0.209	-0.209	GSE292700	40.1%
12	10	73248	12	25437	388	0	0.389	0.174	0.067	0.334	-0.215	-0.215	GSE292700	45.1%

10. Recommended decision

Provisional core: hold all 461 cells for review rather than freezing them as training NK truth. Cluster 19 remains the strongest NK-like region and contains 96.1% of all primary NK anchors, but the new TCR constant-chain audit exposes possible T-cell dropout, ambient RNA, or doublet contamination. Continue to exclude the remaining 8 current calls outside cluster 19.

No automatic expansion is recommended. The next defensible step is a source- and library-resolved audit of TCR UMI depth, ambient burden, doublet scores, and paired TCR evidence within cluster 19. Cluster-19 single-evidence tiers and cluster 9 remain review-only.

review_tier	n_cells	n_sources	dominant_source	maximum_source_fraction	recommended_use
A_core_both_cluster19	461	4	GSE292700	96.7%	provisional core; source/ambient review
B_cluster19_latent_only_dropout_review	594	5	GSE292700	91.1%	dropout-rescue review only
C_cluster19_expression_only_distance_review	759	3	GSE292700	91.7%	source/distance review only
D_cluster9_secondary_nk_review	257	3	GSE294273	57.6%	secondary-cluster review only
E_current_confident_outside_core_hold	7	1	GSE292700	100.0%	hold out of strict NK training
F_no_nk_assignment	111209	5	GSE292700	46.7%	do not assign NK

Decision boundary

No H5AD or frozen label manifest is changed by this report. A defensible expansion should require coherent NK-lineage expression, low T-lineage and myeloid support, localization to reproducible unsupervised clusters, and acceptable source composition. Cytotoxicity alone, one receptor, or the historical scANVI label alone is insufficient.

Reproducibility

- Latent SHA-256: 7030b28be885b85a4b24ce2223bb16f7c7288a29195f2512bf7bbba330791ccf
- Partition SHA-256: 90e83ec83986358a015885e02e29d06eacf726d928e02b2e90428d5c09947a63
- Cluster run: global_r0.8_s29
- Marker-expression cache SHA-256: 82ad58d59e336be57a25cabaa48f56e414f674acdb2657103003ff64c2714225
- Raw CD3/TCR cache SHA-256: 80fadd095a1ed13600ef8fba164b779522a98440de8d28c11b90a7828d43cee5
- Script SHA-256: 4e9cbab26923f63d22f3147ce153bb1582acd046fc785581a82239d15bbaa944
- H5AD mutation: **none**; classifier fitting: **none**; GitHub push: **none**.